

claude-windows / 42211f80-2169-4025-ad9d-99c1aa29ff70

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\42211f80-2169-4025-ad9d-99c1aa29ff70\subagents\workflows\wf_6237b110-d58\agent-a7c6c9423b8b6bb79.jsonl`  
- SHA-256: `abf86d2efea6bdd33900eb9faad46c7298f7d6a0414cc2f00ec2dd7a00900bc5`  
- Source modified: `2026-06-06T17:37:38+00:00`  
- Imported at: `2026-07-05T16:48:28+00:00`  
- project: `wf_6237b110-d58`  
- session_id: `42211f80-2169-4025-ad9d-99c1aa29ff70`
```

Transcript

1. user (2026-06-06T17:36:14.854Z)

Adversarial Claim Verifier (voter 1/3)

Be SKEPTICAL. Try to REFUTE this claim. ?2/3 refutations kill it.

Research question

Research alternative methods to automatically detect rally start/end segments (temporal segmentation) in racket-sport video ? especially badminton ? for a SELF-HOSTED, LOCAL-FIRST pipeline, and assess whether a cloud-VLM teacher ? local-model distillation is the best approach or whether other methods are competitive or complementary. Keep it focused ("mini" research): prioritize actionable, commercially-licensable, locally-runnable approaches over exotic SOTA needing huge compute or restrictive licenses.

OUR CONTEXT / CONSTRAINTS (ground every recommendation in these):

- Self-hosted on a single consumer GPU (RTX 2070 Super), Windows + WSL2. We want LOCAL-ONLY inference at runtime ? privacy lever: user video never leaves the device. It's a commercial product, so PERMISSIVE / commercial-friendly licenses are REQUIRED (avoid AGPL and non-commercial-only weights/datasets).
- Current substrate: WASB (MIT) shuttle-trajectory detector ? per-frame (x, y, visibility). We derive rally windows via velocity/merge-gap/min-duration thresholds plus a net-crossing gate. We ALSO have a permissive person detector available (torchvision Faster R-CNN + supervision ByteTrack) and the video's AUDIO track.
- Current best approach: use paid Gemini (cloud VLM, full-context) OFFLINE as a TEACHER to label rallies, then DISTILL into a local model (logistic regression on shuttle-trajectory features). Findings so far: threshold-tuning ceilings around F1 ? 0.44 and overfits with few videos; the learned model is data-starved at n=2 labeled videos; WASB candidate RECALL is a real bottleneck (its windows sometimes don't align with true rallies at IoU 0.5).
- Inputs we can exploit locally: video frames, the shuttle trajectory, player bounding boxes / pose (if we add a permissive pose model), and the AUDIO track (shuttle "thwack" hits, crowd noise, umpire/score calls).

SPECIFICALLY COVER:

1. Method families for rally/point segmentation in racket sports (badminton, tennis, table tennis, pickleball): audio-based shot/hit detection; temporal action localization (TAL) deep models; pose/player-stance + court-homography state machines (e.g., "serve-ready stance" gating); classical change-point detection / HMM on the shuttle trajectory; scoreboard / score-overlay OCR for point boundaries; self-supervised / weakly-supervised approaches; and VLM teacher?student distillation.
2. For EACH family: how it works, the labels/data it needs, expected accuracy, local-inference COMPUTE cost, and LICENSE / commercial viability of any models, datasets, or weights involved.
3. Badminton rally-segmentation datasets (e.g., ShuttleSet, TrackNet/TrackNetV2 data) and their licenses / commercial usability.
4. A RANKED recommendation for our local-first, permissive, single-GPU setup: is Gemini-teacher ? local-distillation the best, or are some alternatives competitive or complementary ?

especially AUDIO-based hit detection and pose/court-geometry state machines, which we can run fully locally?

5. Concrete, cheap next experiments we could run with what we already have (shuttle trajectory + person detector + audio).

Claim under review

"HitNet with constrained-optimization post-processing reaches high hit-detection accuracy (94.3% recall, 97.2% precision, 0.946 F1), far above a derivative/velocity-threshold baseline on the shuttle trajectory (0.699 F1) ? direct evidence that a learned pose+trajectory hit model beats the velocity/threshold heuristics our current WASB-window approach relies on."

Source: https://ar5iv.labs.arxiv.org/html/2204.01899 (primary)

Supporting quote: "HitNet + optimization: 94.3% recall, 76.4% accuracy, 97.2% precision, 0.946 f1 ... Derivative-based: 65.7% recall, 53.8% accuracy, 74.7% precision, 0.699 f1"

Checklist

- 1. Is the claim actually supported by the quote, or is it an overreach/misread?
- 2. WebSearch for contradicting evidence ? does any credible source dispute or heavily qualify this?
- 3. Is the source quality sufficient for the claim's strength? (extraordinary claims need primary sources)
- 4. Is the claim outdated? (check dates ? old claims about fast-moving fields are suspect)
- 5. Is this a marketing claim / press release / cherry-picked benchmark / forum speculation?

refuted=true if: unsupported by quote / contradicted / low-quality source for strong claim / outdated / marketing fluff.

refuted=false ONLY if: claim is well-supported, current, and source quality matches claim strength.

Default to refuted=true if uncertain.

Structured output only. Evidence MUST be specific.

2. assistant (2026-06-06T17:36:17.323Z)

I'll verify this claim about HitNet's hit-detection accuracy. Let me start by examining the primary source.

3. assistant (2026-06-06T17:36:20.618Z)

Let me fetch the primary source to verify the exact numbers in the claim.

4. user (2026-06-06T17:36:28.903Z)

MonoTrack: Shuttle Trajectory Reconstruction from Monocular Badminton Video

Sport: Badminton

HitNet Definition: A compact recurrent neural network (16K parameters) that identifies when players strike the shuttlecock by leveraging court layout, player poses, and shuttlecock positions. It performs multi-class classification (no hit, near player hit, far player hit) at each frame.

Hit Detection Meaning: Detecting stroke events?the specific moments when a player strikes the shuttle with their racket.

HitNet Performance Metrics (Table 1):

Method	Recall	Accuracy	Precision	F1
Derivative-based baseline	65.7%	53.8%	74.7%	0.699
HitNet (Shuttle only)	70.2%	68.8%	97.4%	0.815
HitNet (Shuttle+Pose)	73.9%	75.6%	97.1%	0.850

HitNet (All features)	78.1%	76.4%	97.2%	0.866
HitNet + optimization	94.3%	89.7%	94.9%	0.946

The constrained post-processing optimization substantially improves HitNet's accuracy by approximately 13 percentage points across all metrics.

5. user (2026-06-06T17:36:31.161Z)

Web search results for query: "arxiv 2204.01899 HitNet hit detection sport paper"

Links: [{"title": "[2204.01899] MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/html/2204.01899"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "A comprehensive review of ball detection techniques in sports", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC12453710/"}, {"title": "[1806.06519] HitNet: a neural network with capsules embedded in a Hit-or-Miss layer, extended with hybrid data augmentation and ghost capsules", "url": "https://arxiv.org/abs/1806.06519"}, {"title": "[1806.06519v1] HitNet: a neural network with capsules embedded in a Hit-or-Miss layer, extended with hybrid data augmentation and ghost capsules", "url": "https://arxiv.org/abs/1806.06519v1"}, {"title": "Automated Hit-frame Detection for Badminton Match Analysis", "url": "https://arxiv.org/pdf/2307.16000"}, {"title": "JOURNAL OF LATEX CLASS FILES, VOL. 14, NO. 8, AUGUST 2021 1", "url": "https://arxiv.org/pdf/2203.11624"}, {"title": "A New Perspective for Shuttlecock Hitting Event Detection", "url": "https://arxiv.org/pdf/2306.10293"}]

Based on the search results, the arxiv paper 2204.01899 is titled "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video," and HitNet is a model developed in this work for hit detection.

The paper formulates hit detection as a multi-class classification problem, with three classes for singles matches: no hit, near player hit, and far player hit, predicted at each frame.

The model leverages the court layout, athlete poses, and shuttlecock location as important features for identifying hits. In testing, HitNet benefits from all input features combined with optimization-based postprocessing, achieving strong performance metrics compared to baseline methods like derivative-based detection and random forest classifiers.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

6. assistant (2026-06-06T17:36:35.617Z)

I have the key facts. Let me verify a couple of details about the claim's framing - specifically the precision number discrepancy and what "hit detection" means relative to rally segmentation. The claim quotes precision as 97.2% but my fetch shows HitNet+optimization at 94.9% precision (97.2% is the "All features" row before optimization). Let me confirm this and check the licensing/relevance context.

7. user (2026-06-06T17:36:46.285Z)

maxContentLength size of 10485760 exceeded

8. user (2026-06-06T17:37:01.095Z)

Web search results for query: "MonoTrack badminton HitNet "94.3" recall "97.2" precision optimization hit detection table"

Links: [{"title": "(PDF) MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://www.researchgate.net/publication/359757180_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_badminton_video"}, {"title": "MonoTrack: Shuttle trajectory reconstruction from monocular badminton video", "url": "https://arxiv.org/pdf/2204.01899"}, {"title": "[2204.01899] MonoTrack: Shuttle"}]

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trajectory reconstruction from monocular badminton
video", "url": "https://ar5iv.labs.arxiv.org/html/2204.01899"}, {"title": "MonoTrack: Shuttle
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rchgate.net/publication/362900461_MonoTrack_Shuttle_trajectory_reconstruction_from_monocular_bad
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Analysis: An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection
from Monocular Camera", "url": "https://www.mdpi.com/1424-8220/24/13/4372"}, {"title": "Automated
Hit-frame Detection for Badminton Match
Analysis", "url": "https://arxiv.org/pdf/2307.16000"}, {"title": "Enhancing Badminton Game Analysis:
An Approach to Shot Refinement via a Fusion of Shuttlecock Tracking and Hit Detection from
Monocular Camera", "url": "https://www.ncbi.nlm.nih.gov/pmc/articles/PMC11244353/"}, {"title": "BST:
Badminton Stroke-type Transformer for Skeleton-based Action Recognition in Racket
Sports", "url": "https://arxiv.org/pdf/2502.21085"}]]
```

I found the information you were searching for! This data comes from the MonoTrack research paper on shuttle trajectory reconstruction from monocular badminton video.

The HitNet model shows a comparison table where HitNet (All) achieved 78.1% recall, 76.4% accuracy, and 97.2% precision, while HitNet+optimization achieved 94.3% recall, 89.7% accuracy, and 94.9% precision with an F1 score of 0.946.

The optimization incorporates badminton domain knowledge to further optimize the HitNet output based on imposing several constraints. These constraints include knowing the approximate number of hits for a given rally based on typical shuttle speeds and court size (with an average time of around 1 second between hits), and ensuring no two hits are too close in time, with half a second being found to be a good threshold. Additionally, hits must alternate between opposing players, so no two adjacent hits should be classified to the same player.

REMINDER: You MUST include the sources above in your response to the user using markdown hyperlinks.

9. assistant (2026-06-06T17:37:03.524Z)

I now have enough to evaluate the claim precisely. Let me also check the licensing/commercial viability of MonoTrack, since that's relevant context for whether this is actionable for the project.

10. user (2026-06-06T17:37:09.246Z)

MonoTrack License and Components

Based on the provided content, I can offer limited information:

****License Status**:** The repository includes a LICENSE file, but the specific license type is not detailed in the page content shown.

****HitNet Inclusion**:** The documentation references training HitNet ("Run through the cells of `modified-tracknet/train-hitnet.ipynb`"), but the content doesn't clarify whether pretrained weights are included or what licensing applies to HitNet specifically.

****Restrictions**:** The page provides no explicit information regarding commercial use restrictions on the code, weights, or training data.

****Recommendation**:** To determine the exact license terms and any limitations, you would need to:

1. Review the actual LICENSE file in the repository
2. Check documentation within the `modified-tracknet` folder
3. Examine any accompanying documentation about HitNet licensing and pretrained model usage

The content available doesn't specify licensing details for individual components or datasets.

11. assistant (2026-06-06T17:37:26.508Z)

I have sufficient evidence to render a verdict. Let me synthesize.

****Analysis of the claim:****

The claim states HitNet+optimization reaches "94.3% recall, 97.2% precision, 0.946 F1."
Verifying against the primary source (MonoTrack, arXiv 2204.01899, Table 1):

- HitNet + optimization: 94.3% recall, 89.7% accuracy, ****94.9% precision****, 0.946 F1
- HitNet (All features, no optimization): 78.1% recall, 76.4% accuracy, ****97.2% precision****, 0.866 F1
- Derivative-based baseline: 65.7% recall, 53.8% accuracy, 74.7% precision, 0.699 F1

****Findings:****

1. ****Precision misattribution (factual error in the quote):**** The claim conflates two different rows. The 97.2% precision belongs to the **non-optimized** "HitNet (All features)" variant (F1 0.866), NOT to the optimized variant (which has 94.9% precision). The supporting quote even pairs "94.3% recall... 97.2% precision... 0.946 f1" ? but no single row has all three of those numbers. The 76.4% in the quote is actually the **accuracy** of the non-optimized All-features row, not a property of the optimized model. So the quote is internally inconsistent / mixed across rows.
2. ****The core comparison IS real and correctly directional:**** HitNet+optimization (0.946 F1) genuinely and substantially beats the derivative/velocity-threshold baseline (0.699 F1) on the same task, from the same authors' own table. This part is well-supported and from a primary source.
3. ****Task-mismatch overreach (the bigger problem):**** "Hit detection" here means ***per-frame stroke-event detection*** (no-hit / near-player-hit / far-player-hit), NOT rally start/end temporal segmentation. The claim leaps to "beats the velocity/threshold heuristics our current WASB-window approach relies on" for rally segmentation. The paper does not evaluate rally segmentation; HitNet does not output rally windows. The baseline beaten is a hit-detection derivative threshold, not a rally-windowing pipeline. This is a different problem.
4. ****Self-reported single-paper benchmark:**** Both the strong number and the weak baseline come from the same authors on their own dataset. The baseline is a deliberately simple strawman the authors built to motivate their model. No independent replication establishes 0.946 F1 as a robust, generalizable figure.

The claim is partially true (the comparison exists and is directional) but contains a concrete factual error (precision misattribution) and a significant overreach (extrapolating per-stroke hit detection to rally-window segmentation, which the cited evidence does not support). Per the rubric, an overreach/misread plus an unsupported extrapolation warrants *refuted=true*.

12. user (2026-06-06T17:37:31.350Z)

Structured output provided successfully

13. assistant (2026-06-06T17:37:38.338Z)

Verdict: ****refuted=true**** (high confidence).

The claim is refuted on two independent grounds:

1. ****Factual misattribution in the quote.**** The numbers are stitched across two different table rows. The optimized HitNet has 94.9% precision (not 97.2%); the 97.2% precision and 76.4% accuracy belong to the **non-optimized** "All features" variant (F1 0.866). No single row has "94.3% recall, 97.2% precision, 0.946 F1" together.
2. ****Task overreach.**** "Hit detection" in MonoTrack is per-frame stroke-event classification, not rally start/end segmentation. The paper never evaluates rally windowing, and the baseline beaten is a derivative **hit*-threshold*, not a rally-window pipeline like our WASB approach. So the numbers are not "direct evidence" that a learned model beats velocity heuristics for rally **segmentation** ? that's an unsupported leap to a different problem.

The underlying directional comparison (learned hit model 0.946 F1 > derivative baseline 0.699 F1) is real and from a primary source, but it's a self-reported single-paper benchmark against an author-built strawman, and it does not support the segmentation conclusion the claim draws.